

2026 World Cup Prediction Using Monte Carlo Simulation

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THE PREDICTION OF 2026 WORLDCUP USING MONTE CARLO

Load library

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.3
```

```
##  
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':  
##  
##   filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(tidyr)
```

1. READ & CLEAN DATA

```
#importing historical match results data  
data_games <- read.csv("results.csv", stringsAsFactors = FALSE)  
data_games$date <- as.Date(data_games$date)  
  
#using more recent data (e.g., since 2015) for better performance  
data_modern <- data_games %>% filter(date >= "2015-01-01")  
  
#calculating the overall average number of goals  
avg_home_goals <- mean(data_modern$home_score, na.rm = TRUE)  
avg_away_goals <- mean(data_modern$away_score, na.rm = TRUE)
```

2. CALCULATING PERFORMANCE TEAM (ATTACK & DEFENSE RATINGS)

```

#home_stats
home_stats <- data_modern %>%
  group_by(home_team) %>%
  summarise(
    games_home = n(),
    goals_scored_home = sum(home_score, na.rm = TRUE),
    goals_conceded_home = sum(away_score, na.rm = TRUE)
  ) %>%
  rename(team = home_team)

#away_stats
away_stats <- data_modern %>%
  group_by(away_team) %>%
  summarise(
    games_away = n(),
    goals_scored_away = sum(away_score, na.rm = TRUE),
    goals_conceded_away = sum(home_score, na.rm = TRUE)
  ) %>%
  rename(team = away_team)

#combining stats and calculating team strength ratings
team_ratings <- full_join(home_stats, away_stats, by = "team") %>%
  mutate(
    games_home = replace_na(games_home, 0),
    games_away = replace_na(games_away, 0),
    goals_scored_home = replace_na(goals_scored_home, 0),
    goals_conceded_home = replace_na(goals_conceded_home, 0),
    goals_scored_away = replace_na(goals_scored_away, 0),
    goals_conceded_away = replace_na(goals_conceded_away, 0),

    total_games = games_home + games_away,
    #counting attack and defense rating globally
    attack_rating = ((goals_scored_home + goals_scored_away) / total_games) /
      ((avg_home_goals + avg_away_goals)/2),
    defense_rating = ((goals_conceded_home + goals_conceded_away) / total_games) /
      ((avg_home_goals + avg_away_goals)/2)
  ) %>%
  filter(total_games >= 5) %>% #filtering team
  select(team, attack_rating, defense_rating)

```

3. Single Match Simulation Function (Knockout Stage)

```

simulasi_match <- function(team_A, team_B, avg_goals) {
#retrieve team ratings; if a team is not listed, assign a default rating of 1.0
att_A <- ifelse(team_A %in% team_ratings$team,
                team_ratings$attack_rating[team_ratings$team == team_A], 1.0)
def_A <- ifelse(team_A %in% team_ratings$team,
                team_ratings$defense_rating[team_ratings$team == team_A], 1.0)
att_B <- ifelse(team_B %in% team_ratings$team,
                team_ratings$attack_rating[team_ratings$team == team_B], 1.0)
def_B <- ifelse(team_B %in% team_ratings$team,
                team_ratings$defense_rating[team_ratings$team == team_B], 1.0)

#counting gol expectation (Lambda Poisson)
lambda_A <- att_A * def_B * avg_goals
lambda_B <- att_B * def_A * avg_goals

#score simulatuon by poisson distribution
score_A <- rpois(1, lambda_A)
score_B <- rpois(1, lambda_B)

if (score_A > score_B) {
  return(team_A)
} else if (score_B > score_A) {
  return(team_B)
} else {
  #if draw --> penalty (50:50 probability)
  return(sample(c(team_A, team_B), 1))
}
}

```

4. SIMULATION OF MONTE CARLO

```

#round of 16
teams_16 <- c(
  "Canada", "Morroco", "Paraguay", "France",
  "United States", "Belgium", "Portugal", "Spain",
  "Brazil", "Norway", "Mexico", "England",
  "Switzerland", "Colombia", "Argentina", "Egypt"
)

avg_global_goals <- (avg_home_goals + avg_away_goals) / 2
n_simulations <- 10000
champions <- character(n_simulations)

set.seed(42)
champions <- character(n_simulations)

for (i in 1:n_simulations) {
  # ---- ROUND OF 16 -> QUARTER FINAL ----
  qf1 <- simulasi_match(teams_16[1], teams_16[2], avg_global_goals)
  qf2 <- simulasi_match(teams_16[3], teams_16[4], avg_global_goals)
  qf3 <- simulasi_match(teams_16[5], teams_16[6], avg_global_goals)
  qf4 <- simulasi_match(teams_16[7], teams_16[8], avg_global_goals)
  qf5 <- simulasi_match(teams_16[9], teams_16[10], avg_global_goals)
  qf6 <- simulasi_match(teams_16[11], teams_16[12], avg_global_goals)
  qf7 <- simulasi_match(teams_16[13], teams_16[14], avg_global_goals)
  qf8 <- simulasi_match(teams_16[15], teams_16[16], avg_global_goals)

  # ---- QUARTER FINAL -> SEMI FINAL ----
  sf1 <- simulasi_match(qf1, qf2, avg_global_goals)
  sf2 <- simulasi_match(qf3, qf4, avg_global_goals)
  sf3 <- simulasi_match(qf5, qf6, avg_global_goals)
  sf4 <- simulasi_match(qf7, qf8, avg_global_goals)

  # ---- SEMI FINAL -> FINAL ----
  f1 <- simulasi_match(sf1, sf2, avg_global_goals)
  f2 <- simulasi_match(sf3, sf4, avg_global_goals)

  # ---- FINAL -> CHAMPION ----
  champions[i] <- simulasi_match(f1, f2, avg_global_goals)
}

```

5. THE RESULT OF SIMULATION

```

results_table <- table(champions) %>%
  as.data.frame() %>%
  rename(Country = champions, Champs_Freq = Freq) %>%
  mutate(Champs_Probability = paste0(round((Champs_Freq / n_simulations) *
                                          100, 2), "%")) %>%
  arrange(desc(Champs_Freq))

print("--- 2026 World Cup Prediction Using Monte Carlo Simulation ---")

```

```
## [1] "--- 2026 World Cup Prediction Using Monte Carlo Simulation ---"
```

```
print(results_table)
```

##	Country	Champs_Freq	Champs_Probability
## 1	Brazil	1201	12.01%
## 2	England	1180	11.8%
## 3	Argentina	1164	11.64%
## 4	France	1111	11.11%
## 5	Spain	1076	10.76%
## 6	Belgium	908	9.08%
## 7	Portugal	883	8.83%
## 8	Canada	664	6.64%
## 9	Egypt	450	4.5%
## 10	Colombia	336	3.36%
## 11	Switzerland	274	2.74%
## 12	United States	271	2.71%
## 13	Mexico	218	2.18%
## 14	Norway	197	1.97%
## 15	Morroco	51	0.51%
## 16	Paraguay	16	0.16%